Singapore School Kinderland

Secondary 1 (Sec 1) — Class Test 2 (Term 2) — TEACHER ANSWER KEY

Subject: Computer Science | Chapter 2.3: Encryption & Security (TB Pg 63–74) | Max Marks: 25

Candidate Name: _____	**Class / Sec: Sec 1**	**Roll No:** _____
Date: _____	**Time Allowed: 45 Mins**	**Marks Obtained:** MARKING SCHEME

INSTRUCTIONS: Answer ALL questions in the spaces provided. Write concise, accurate technical descriptions using Cambridge Computer Science terminology. No calculators permitted.

Question 1: Principles of Encryption & Terminology *[5 Marks]*

(a) Define the following fundamental encryption terms: *[3]*

(i) **Plaintext:**

(ii) **Ciphertext:**

(iii) **Encryption Key:**

(b) A Caesar shift cipher replaces each letter with the letter 3 positions ahead in the alphabet (e.g. A → D). Encrypt the word **"SECURITY"** using this algorithm: *[2]*

Encrypted Ciphertext: **V H F X U L W B** (S→V, E→H, C→F, U→X, R→U, I→L, T→W, Y→B)

Question 2: Symmetric Encryption & Security Challenges *[5 Marks]*

(a) State the defining characteristic of Symmetric Encryption: *[1]*

(b) Explain what is meant by the **Key Distribution Problem** and why it is a major security vulnerability when transmitting data over the Internet: *[2]*

(c) Define a **Brute-Force Attack** and explain how key length (e.g. 256-bit vs 56-bit) defends against it: *[2]*

Question 3: Asymmetric Encryption & Key Pair Mechanics ***[6 Marks]***

Asymmetric encryption uses a mathematically linked pair of keys: a **Public Key** and a **Private Key**.

(a) State the difference in accessibility between a user's Public Key and their Private Key: ***[2]***

(b) **Scenario:** Doctor Sarah wishes to send confidential medical records to Hospital B. Describe the exact sequence of keys used to encrypt and decrypt the records: ***[2]***

Key used by Dr. Sarah to encrypt: **Hospital B's Public Key** [1]

Key used by Hospital B to decrypt: **Hospital B's Private Key** [1]

(c) State ONE operational advantage and ONE operational disadvantage of Asymmetric Encryption compared to Symmetric Encryption: ***[2]***

Advantage:

Disadvantage:

Question 4: Digital Signatures & Non-Repudiation ***[5 Marks]***

(a) Describe how a **Digital Signature** is generated by a sender and verified by a recipient: ***[3]***

(b) Explain why a Digital Signature provides **Non-Repudiation** (the sender cannot deny sending it): ***[2]***

Question 5: SSL/TLS & Digital Certificates ***[4 Marks]***

(a) State the role of a trusted **Certificate Authority (CA)** in web security: ***[1]***

(b) List TWO items stored inside a website's SSL/TLS Digital Certificate: ***[2]***

Item 1: _____

Item 2: _____

(c) State what visual indicator appears in a web browser to confirm an active SSL/TLS encrypted connection: ***[1]***